A Secure File Sharing System Based on IPFS and Blockchain

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by

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(AN ISO 9001:2008 CERTIFIED INSTITUTION)
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Certificate

This is to certify that the seminar report titled **A Secure File Sharing System Based on IPFS and Blockchain** is a bonafide record of the work carried out by **Mohammed Nabeel A A (Univ.Reg.No. VAS23CS134)** of Vidya Academy of Science & Technology, Thalakkottukara, Thrissur - 680 501 in partial fulfillment of the requirements for the award of **Degree of Bachelor of Technology** in **Computer Science and Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2023-2027.

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Abstract

Blockchain provides decentralization, immutability and denial-proof transaction tracking, whereas the InterPlanetary File System (IPFS) provides cost-efficient, content-addressable, peer-to-peer storage. Combining the two is therefore an attractive way to build file-sharing systems, since encrypted file payloads can be kept on IPFS while identifiers and verification hashes are anchored on the blockchain. However, existing integrations still suffer from heavy client-side computation, risky management of raw sub-keys at client nodes, absence of native access control in IPFS, and expensive user revocation. In this seminar, a secure file sharing system based on IPFS and blockchain with decentralized access control and group key management is studied. The architecture introduces a dedicated *IPFS Proxy* that generates symmetric group keys, encrypts files, maintains group mapping tables and performs key wrapping with the public keys of authorized members, so that the file owner is relieved from heavy computation. When a user is revoked, the proxy automatically generates a fresh group key, re-encrypts all group files, anchors the new IPFS hashes on the blockchain and wraps the new key only for the remaining users, which gives a communication complexity of $O(n)$ for a group of n active users. A comparison with state-of-the-art schemes shows that the proposed model achieves extremely low owner computation cost, minimal client security risk, denial-proof integrity and seamless dynamic group support. The current prototype is being implemented with Hyperledger Fabric and local IPFS testnets, and detailed latency benchmarks are planned as future work.

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Vision of the Institution

To be a premier institution in engineering education and research, nurturing innovative professionals for sustainable societal development.

Mission of the Institution

M1. Provide quality engineering education that empowers students with strong technical competence and analytical problem solving abilities.

M2. Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable technological solutions.

M3. Develop ethical and socially responsible professionals with leadership qualities, lifelong learning abilities and a commitment to societal well-being.

Vision of the Department

To excel in computer science and engineering education and research, nurturing innovative and ethically responsible professionals for sustainable technological development.

Mission of the Department

M1. Impart quality education in computer science and engineering by providing strong theoretical foundations, practical skills and exposure to emerging technologies.

M2. Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable and socially relevant computing solutions.

M3. Develop graduates with professional competence, ethical values, logical and analytical thinking skills, leadership qualities and a commitment to lifelong learning and societal well-being.

Program Educational Objectives (PEOs)

PEO-1: Graduates shall practice computer science and engineering in IT industry, public sector undertaking, as an entrepreneur or as a researcher pursuing a successful professional career.

PEO-2: Graduates shall demonstrate creativity, develop innovative ideas and work in teams to accomplish a common goal.

PEO-3: Graduates shall demonstrate project management skills, lifelong learning and social awareness upholding ethical values.

Program Outcomes (POs)

PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health, safety, and environment. (WK5)
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations using research-based knowledge including design of experiments, modelling, analysis and interpretation of data. (WK8)
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques and modern engineering tools recognizing their limitations. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze societal and environmental impacts with reference to sustainability, economy, health, safety, and culture. (WK1, WK5, WK7)
PO7	**Ethics:** Apply ethical principles, professional ethics, and adhere to laws. (WK9)
PO8	**Individual and Collaborative Team Work:** Function effectively as an individual and in teams.

PO9	**Communication:** Communicate effectively through reports, documentation, and presentations considering diverse audiences.
PO10	**Project Management and Finance:** Apply management principles and economic decision-making in projects.
PO11	**Life-Long Learning:** Recognize the need for independent and lifelong learning. (WK8)

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Specific Outcomes (PSOs)

PSO-1	The ability to apply the knowledge of computational structures and algorithms for solving complex engineering problems in multidisciplinary areas.
PSO-2	The ability to design, develop and evaluate computer-based systems using advanced tools and technologies to meet real-world challenges.

Sustainable Development Goals (SDGs)

SDG 1	**No Poverty:** End poverty in all its forms everywhere.
SDG 2	**Zero Hunger:** End hunger, achieve food security and improved nutrition and promote sustainable agriculture.
SDG 3	**Good Health and Well-being:** Ensure healthy lives and promote well-being for all at all ages.
SDG 4	**Quality Education:** Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.
SDG 5	**Gender Equality:** Achieve gender equality and empower all women and girls.
SDG 6	**Clean Water and Sanitation:** Ensure availability and sustainable management of water and sanitation for all.
SDG 7	**Affordable and Clean Energy:** Ensure access to affordable, reliable, sustainable and modern energy for all.
SDG 8	**Decent Work and Economic Growth:** Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.
SDG 9	**Industry, Innovation and Infrastructure:** Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.
SDG 10	**Reduced Inequalities:** Reduce inequality within and among countries.
SDG 11	**Sustainable Cities and Communities:** Make cities and human settlements inclusive, safe, resilient and sustainable.
SDG 12	**Responsible Consumption and Production:** Ensure sustainable consumption and production patterns.

SDG 13	**Climate Action:** Take urgent action to combat climate change and its impacts.
SDG 14	**Life Below Water:** Conserve and sustainably use the oceans, seas and marine resources for sustainable development.
SDG 15	**Life on Land:** Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.
SDG 16	**Peace, Justice and Strong Institutions:** Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.
SDG 17	**Partnerships for the Goals:** Strengthen the means of implementation and revitalize the global partnership for sustainable development.

SDGs addressed with Justification

Seminar Title	**SDG addressed**	**Justification**
Secure File Sharing using IPFS and Blockchain	SDG 9: Industry, Innovation and Infrastructure	Presents a decentralized file sharing infrastructure built on IPFS, blockchain and an IPFS proxy with group key management, reducing dependence on centralized providers. This is the primary goal addressed.
	SDG 4: Quality Education	Supports secure sharing of academic documents among groups of students and staff with controlled access and cheap revocation.
	SDG 3: Good Health and Well-being	Allows healthcare records to be shared with authorized groups only and withdrawn when no longer needed, protecting patient privacy.

POs & PSOs addressed with Justification

Seminar Title	POs & PSOs addressed	Justification
Secure File Sharing using IPFS and Blockchain	PO1 – Engineering Knowledge	Applies symmetric encryption (AES-CBC), public-key cryptography, hashing, IPFS and blockchain fundamentals to secure file sharing.
	PO2 – Problem Analysis	Analyzes limits of current schemes such as heavy client computation, risky sub-key management, no native IPFS access control and costly revocation.
	PO3 – Design/Development of Solutions	Studies a design with an IPFS proxy, group mapping tables and key wrapping that relieves the file owner of heavy computation.
	PO4 – Conduct Investigations of Complex Problems	Reviews prior sharing schemes and compares owner cost, client risk, integrity and dynamic group support.
	PO5 – Modern Tool Usage	Examines IPFS, Hyperledger Fabric and cryptographic tools, and recognizes limits such as proxy trust and revocation cost.
	PO6 – The Engineer and The World	Gives users control of group access to their data and reduces the impact of data exposure.

Seminar Title	**POs & PSOs addressed**	**Justification**
	PO7 – Ethics	Keeps raw keys away from clients, anchors hashes transparently on the blockchain and respects data owners' privacy.
	PO8 – Individual and Team Work	Works independently and incorporates feedback from the guide, coordinators and peers during the seminar.
	PO9 – Communication	Communicates the study through this report and a seminar presentation.
	PO10 – Project Management and Finance	Plans the seminar within the schedule and considers gas, hosting and infrastructure cost in the design.
	PO11 – Life-Long Learning	Studies emerging technologies independently, drawing on research papers and documentation.
	PSO1	Applies group key generation, key wrapping and re-encryption logic to control who can read a file.
	PSO2	Evaluates a prototype built with IPFS and Hyperledger Fabric against real file sharing needs.

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List of Abbreviations

IPFS	InterPlanetary File System
P2P	Peer-to-Peer
AES	Advanced Encryption Standard
CBC	Cipher Block Chaining
PK	Public Key
IoT	Internet of Things
CID	Content Identifier
HLF	Hyperledger Fabric
DApp	Decentralized Application

Chapter 1

INTRODUCTION

1.1 Background and Motivation

In recent years, the volume of digital data that individuals and organizations store and exchange has grown enormously, and sharing files securely among a dynamic set of collaborators has become a central requirement of modern computing. Traditional cloud storage places complete trust in a single provider, which creates a single point of failure, exposes data to insider threats and makes the stored content difficult to audit. Decentralized technologies such as blockchain and peer-to-peer storage have therefore attracted considerable attention as an alternative foundation for secure file sharing.

Blockchain is a decentralized and tamper-proof ledger whose built-in immutability guarantees the honesty of transactions. Linked cryptographic blocks make every transaction traceable and completely denial-proof. However, a blockchain is a highly expensive and resource-limited database, which makes it unsuitable for storing large files directly. The InterPlanetary File System (IPFS), on the other hand, locates files through cryptographically unique hashes rather than IP-based URLs and offers a cost-efficient and highly scalable peer-to-peer block storage mechanism. Its limitation is that content is publicly accessible by nature and IPFS provides neither access policies nor integrated encryption.

To mitigate these limitations, there is a need for a scheme that stores the *encrypted file payloads on IPFS* and anchors the *identifications and verification hashes on the blockchain* for immutable tracking. Such a hybrid design requires an efficient key management and access control mechanism so that only authorized group members can decrypt the shared files, even though the storage layer itself is open.

1.2 Technological Advancements in Secure File Sharing

Technological progress in distributed ledgers and content-addressable storage has transformed the way data is shared. Public and permissioned blockchains such as Ethereum and Hyperledger Fabric allow smart contracts to record metadata, ownership and access events in an auditable manner, while IPFS removes the dependence on a central server by distributing file blocks across many nodes.

Several researchers have proposed blockchain-assisted file-sharing frameworks. For IoT data, schemes based on proxy re-encryption tokens assign key generation and data updates entirely to the data owner, which overwhelms thin IoT nodes. Distributed key schemes spread sub-keys among several parties to achieve scalable file sharing, but they force client nodes to manage raw sub-keys manually and thereby increase the risk of key leakage and malware attacks.

Symmetric ciphers such as AES in CBC mode remain the most efficient choice for encrypting bulk file content, while public-key cryptography is used for authentication, digital signatures and for protecting (wrapping) the symmetric keys. The combination of symmetric file encryption with public-key key wrapping forms the cryptographic backbone of the system studied in this seminar.

1.3 Role of IPFS and Blockchain in Decentralized Storage

In IPFS a file is split into blocks and every block is addressed by a cryptographic hash (content identifier). Any change to the content changes its address, which provides built-in integrity checking. Because IPFS is permission-less, anyone who knows the hash can retrieve the data; confidentiality therefore has to be provided by encrypting the file *before* it is stored.

The blockchain complements IPFS by acting as a trustworthy index. It stores the metadata of every uploaded file, such as the owner identity, group identifier, file hash and the IPFS hash, inside a transaction identified by a unique `trans_id`. Authorized users query the blockchain with this transaction identifier to obtain the file details and the IPFS hash, and can verify that the retrieved content has not been modified. By delegating heavy cryptographic operations to a dedicated IPFS Proxy, the combined system avoids burdening the owner and the clients while preserving decentralization.

1.4 Goals and Purposes

The overarching goal of this seminar is to study and present a secure file sharing system that combines IPFS and blockchain with decentralized access control and efficient group key management. The scheme aims to eliminate the operational gaps present in existing integrations, namely heavy client computation, critical security gaps, lack of native access control and complex user revocation.

Another primary goal is to reduce the computational and storage overhead on the file owner and on the group users by delegating key generation, file encryption, key wrapping and re-encryption to an independent IPFS Proxy, without placing unconditional trust in any central authority. Every critical operation is anchored on the blockchain so that integrity can be proven later.

Ultimately, the purpose of this study is to demonstrate how fine-grained, dynamic group access can be enforced on an otherwise open storage network, how users can be revoked with a communication overhead of $O(n)$, and to lay a foundation for scalable, collaborative and trustworthy decentralized file sharing.

Chapter 2

LITERATURE REVIEW

2.1 LITERATURE SURVEY

2.1.1 IPFS - Content Addressed, Versioned, P2P File System

- **Authors:** Benet, 2014
- **Source:** arXiv preprint (arXiv:1407.3561)

Benet (2014) introduced IPFS as a peer-to-peer distributed file system that connects all computing devices with the same system of files. IPFS combines a distributed hash table, an incentivized block exchange and a self-certifying namespace, and organizes data as a Merkle directed acyclic graph in which every object is addressed by the hash of its content. This content addressing provides tamper resistance, de-duplication and efficient distribution.

The system is highly scalable and cost-efficient for large files. Nevertheless, IPFS itself does not provide access control or encryption, so every file whose hash is known can be retrieved by any peer. This motivates the use of external encryption and key management layers on top of IPFS, which is the gap addressed by the proxy-based architecture in this seminar.

2.2 LITERATURE SURVEY

2.2.1 Blockchain-Based Data Sharing Scheme for IoT with Re-encryption Tokens

- **Authors:** *[Add authors of reference [3]]*

- **Source:** *[Add journal / conference and year]*

This scheme integrates blockchain with IPFS for IoT data sharing and uses re-encryption tokens to grant access to data users. Metadata is anchored on the blockchain while the encrypted data is held in IPFS. Integrity proof is mainly based on hash matching.

The approach demonstrates that blockchain can support auditable sharing of IoT data. However, key generation and data updates are assigned entirely to the data owner, so the owner computation cost is extremely high, which overwhelms thin IoT nodes. User revocation must also be handled by the owner, and support for dynamic groups is limited.

2.3 LITERATURE SURVEY

2.3.1 Scalable File Sharing Using Distributed Sub-Keys over Blockchain and IPFS

- **Authors:** *[Add authors of reference [5]]*
- **Source:** *[Add journal / conference and year]*

This work proposes a scalable file sharing scheme in which group keys are split into sub-keys that are distributed among the participants, with the blockchain providing a trustworthy record of operations. The owner computation is reduced to a medium level through key sharing, and the integrity record is trustworthy.

The main limitation is that client nodes have to store and manage raw sub-keys, which drastically increases the leakage risk and exposure to malware attacks. Supporting dynamic groups is highly complex, and revocation causes significant storage and execution latency. The proxy-based scheme studied in this seminar was proposed to overcome these drawbacks.

Chapter 3

COMPONENTS OF THIS SYSTEM

3.1 File Owner and IPFS Proxy

The **File Owner** registers groups, signs the data, uploads files and governs the membership policies of its groups. The **IPFS Proxy** is the core engine of the architecture: it generates symmetric group keys, verifies owner signatures, encrypts files, uploads them to IPFS, maintains the mapping tables and executes key wrapping. Delegating these operations to the proxy removes the heavy computation from the owner.

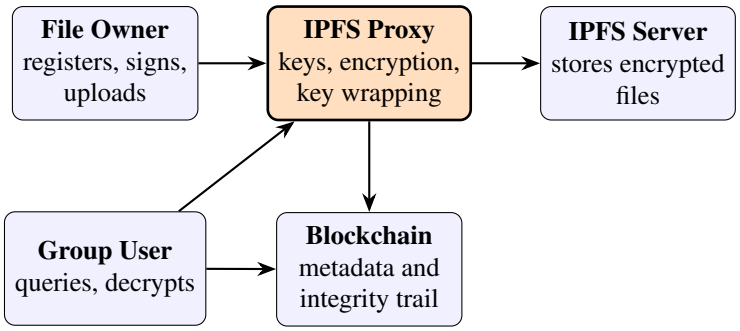


Figure 3.1: Core entities of the proposed system

3.2 IPFS Server and Blockchain

The **IPFS Server** accepts and distributes the cryptographically encrypted files and returns a unique file hash (`IPFS_hash`) for every stored file. Since only ciphertext is stored, public reachability of the IPFS network does not expose the content.

The **Blockchain** validates metadata and transactions and creates an undeniable integrity trail. For every upload, the proxy anchors the metadata (owner, group, file hash and `IPFS_hash`) in a transaction and receives a `trans_id`, which is relayed to the owner. The prototype uses Hyperledger Fabric as the permissioned ledger.

3.3 Group User and Proxy Mapping Table

A **Group User** registers its public key at the proxy, obtains the owner's authorization to join a group, queries the blockchain for file details and decrypts files using the group key wrapped with its own public key.

The IPFS Proxy stores a structured mapping table for every group. The symmetric key (for example AES-CBC) of the group is coupled with the verified public keys of its members, as shown in Table 3.1.

Table 3.1: Proxy mapping table structure

Group Name	Member ID	Key Parameter (Stored in Proxy)
Group 1	Group 1 Key	Group1_Symmetric_Key (AES)
Group 1	Alice	Alice_Public_Key
Group 1 & 2	Bob	Bob_Public_Key
Group 2	Dave	Dave_Public_Key

3.4 Group Overlap Model

A user can belong to several groups. In Figure 3.2, Bob belongs to both Group 1 and Group 2, whereas Alice and Carol belong only to Group 1 and Dave only to Group 2. The proxy isolates access through the mapping table so that Alice cannot read Group 2 files and Dave cannot read Group 1 files.

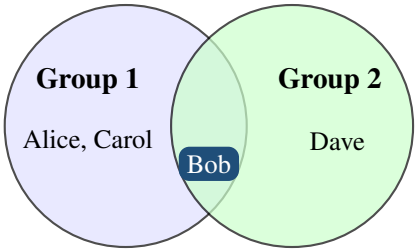


Figure 3.2: Group overlap model

Chapter 4

METHODOLOGY

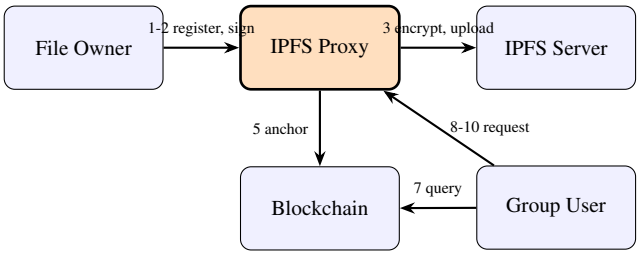


Figure 4.1: Working flow chart

4.1 Working Mechanism

4.1.1 Working Mechanism

The proposed system is a multi-layer ecosystem built around a dedicated IPFS Proxy that delegates access control and group key management without centralized trust overhead. The File Owner, IPFS Proxy, IPFS Server, Blockchain and Group Users interact through a sequence of registration, upload, download and revocation workflows. The proxy performs all heavy computational operations, while the blockchain records integrity-critical metadata. This design relieves thin and IoT nodes from expensive cryptographic work and removes the need for clients to handle raw sub-keys.

4.1.2 Registration and Group Creation

The owner registers its public key (PK) and `user_id` with the IPFS Proxy, which returns a unique `group_id`. The proxy creates a mapping-table entry for the group and generates its symmetric group key (for example AES-CBC). Users later register their own public keys at the proxy and obtain the owner's authorization to be added to the group mapping.

4.1.3 Upload Workflow

Uploading a file proceeds in five steps. (1) *Register:* the owner registers PK and `user_id` and receives `group_id`. (2) *Sign & Submit:* the owner signs the target file and submits the payload to the proxy for integrity checks. (3) *Secure Encrypt:* the proxy verifies the signature, hashes the file, encrypts it with the group key and uploads it to IPFS. (4) *IPFS Store:* the IPFS server stores the encrypted data and returns the unique file hash `IPFS_hash`. (5) *Anchor Chain:* the proxy anchors the metadata on the blockchain and relays the resulting `trans_id` to the owner.

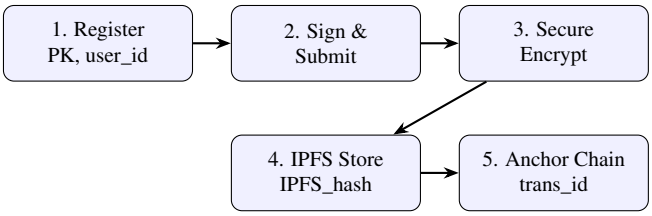


Figure 4.2: Upload workflow

4.1.4 Download Workflow

Authorized users fetch metadata from the blockchain and retrieve the encrypted payload through the proxy using key wrapping. (6) *Register & Permit:* the user registers its key at the proxy and receives the owner's authorization to join the group. (7) *Chain Query:* the user queries the blockchain with `trans_id` to obtain file details and `IPFS_hash`. (8) *Signed Request:* the user signs and sends a download request; the proxy fetches the file from IPFS. (9) *File Fetch:* the IPFS server delivers the encrypted block. (10) *Wrap & Decrypt:* the proxy encrypts the group key with the user's public key (key wrapping); the user unwraps the key, decrypts the file and verifies it against the recorded hash.

4.1.5 User Revocation Mechanism

When a user is revoked from a group, (1) the IPFS Proxy automatically generates a brand-new symmetric group key; (2) the proxy retrieves and re-encrypts all previously shared group files with the updated key; (3) the new encrypted files produce updated IPFS hashes, which

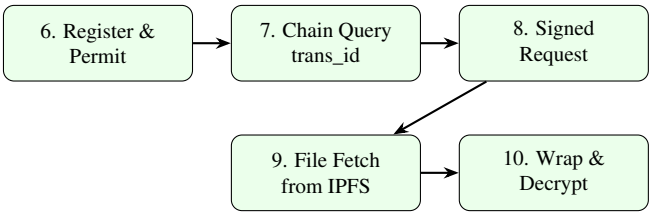


Figure 4.3: Download workflow

are anchored as fresh transactions on the blockchain; and (4) the new group key is wrapped with the public keys of the remaining authorized users only, so the revoked user is fully isolated. Unlike standard architectures, the File Owner does not handle file downloads, re-encryptions or key updates during revocation, which eliminates storage and performance bottlenecks on local client machines.

4.1.6 Communication Overhead Complexity

For a group containing a set of n active users, the new group key has to be wrapped once for every remaining user. The communication complexity of a single revocation therefore executes on the order of

$$O(n),$$

where n represents the number of authorized users in the group. File re-encryption takes place inside the proxy and does not involve the clients.

4.2 Security Comparison

Table 4.1 compares the proposed IPFS Proxy system with two state-of-the-art schemes, the IoT data scheme [3] and the scalable file sharing scheme [5].

Table 4.1: Comparison with state-of-the-art schemes

Feature	IoT Data Scheme [3]	Scalable File Sharing [5]	Proposed IPFS Proxy System
Owner Computation Cost	Extremely High	Medium (Key Sharing)	Extremely Low (Proxy-Delegated)
User Revocation Handler	Handled by Owner	Distributed Sub-Keys	Managed securely by Proxy
Client Security Risks	Medium	High (Clients store sub-keys)	Minimal (Protected via Wrapping)
Integrity Proof	Weak / Hash matching	Trustworthy	Denial-Proof Blockchain Ledger
Dynamic Group Support	Limited	Highly Complex	Seamless Mapping Table Isolation

Chapter 5

APPLICATION

5.1 Secure Collaborative Document Sharing:

Teams can share confidential documents with changing membership. Files remain encrypted on IPFS, access is controlled by group membership in the proxy mapping table, and every upload is traceable on the blockchain.

5.2 Healthcare Record Sharing:

Patient records can be stored as encrypted objects and shared among authorized doctors and hospitals. The immutable audit trail on the blockchain records who uploaded or accessed data, and revocation of a practitioner immediately re-keys the affected records.

5.3 IoT Data Sharing:

Thin IoT nodes can publish sensor data without performing heavy key generation or re-encryption, because these operations are delegated to the proxy. This directly addresses the computation limits of resource-constrained devices.

5.4 Enterprise and Academic Repositories:

Organizations and universities can maintain departmental groups, where a member who belongs to several groups, such as Bob in the overlap model, accesses only the files of his

groups.

5.5 Digital Evidence and Legal Records:

The denial-proof blockchain ledger and hash verification make it possible to prove that a stored document has not been modified, which is valuable for evidence management and legal archives.

5.6 Supply Chain and Contract Management:

Business partners can share contracts and shipment documents inside dynamic groups and revoke a partner at the end of an agreement without any burden on the owner.

5.7 Decentralized Cloud Alternatives:

The architecture offers a trustless alternative to centralized cloud storage, with fine-grained access policies enforced over an open, permission-less storage network.

Chapter 6

CHALLENGES AND LIMITATIONS

6.1 Proxy Trust and Availability

Although the proxy removes the burden from clients, it becomes the engine that holds symmetric group keys. Its compromise or failure could expose keys or interrupt service, so it must be hardened and replicated.

6.2 Revocation Cost for Large Data

Revocation requires the proxy to retrieve and re-encrypt all files previously shared with the group. For groups holding many or very large files this can cause considerable runtime and bandwidth consumption.

6.3 Blockchain Throughput and Cost

Every upload and every re-encryption creates new transactions. Ledger throughput, latency and storage growth limit the scalability of the metadata layer, especially under mass revocation.

6.4 Key Management Security

Secure generation, storage and rotation of AES keys and the correct protection of wrapped keys are critical. Weak randomness or implementation mistakes in CBC mode could com-

promise confidentiality.

6.5 Persistence of Previously Accessed Data

A revoked user who has already downloaded and decrypted a file retains that copy. Re-keying only protects future versions of the data.

6.6 Integration and Interoperability

Combining IPFS nodes, a permissioned blockchain and the proxy requires consistent interfaces, and migration from existing storage systems is non-trivial.

6.7 Cost and Resource Requirements

Operating the proxy, IPFS nodes and ledger peers requires computational and storage resources, and detailed performance benchmarks are still pending.

Chapter 7

FUTURE SCOPE AND DISCUSSION

7.1 Future Developments

7.1.1 Current Prototype Status

The design is currently being implemented using Hyperledger Fabric and local IPFS testnets. The AES-CBC key wrapping workflows have been fully modeled.

7.1.2 Latency Benchmarking

Detailed latency benchmarks under high stress are planned in order to measure upload, download and revocation times for different file sizes and group sizes.

7.1.3 Optimized Re-encryption

Re-encryption runtimes during massive user revocation operations will be optimized, for example through batching, lazy re-encryption or parallel processing in the proxy.

7.1.4 Distributed and Fault-Tolerant Proxy

Multiple proxy nodes with threshold-based key custody can remove the single-proxy dependency and improve availability and trust.

7.1.5 Stronger Cryptographic Primitives

Authenticated encryption modes (such as AES-GCM) and attribute-based or proxy re-encryption techniques can be evaluated to add integrity protection and finer policies.

7.1.6 Smart-Contract Based Policies

Group membership and access policies can be enforced and audited through chaincode, making the mapping decisions verifiable on the ledger.

7.1.7 Integration with IoT and Edge Devices

Lightweight clients for IoT and edge devices can be developed so that constrained nodes can use the system with minimal computation.

7.1.8 Privacy and Compliance

Privacy-preserving metadata and compliance with data protection regulations will be needed for sensitive domains such as healthcare.

7.1.9 Collaborative Platforms and Standardization

The architecture can be extended to collaborative platforms where standards for decentralized access control and group key management are shared across applications.

Chapter 8

CONCLUSION

The present study demonstrates a robust framework for secure file sharing that merges blockchain with IPFS content-addressed storage. By storing encrypted payloads on IPFS and anchoring identifiers and verification hashes on the blockchain, the system overcomes the storage size constraint of blockchains and the lack of confidentiality in IPFS.

The independent IPFS Proxy provides decentralized, fine-grained access control through group mapping tables and key wrapping, and it drastically cuts the computation and storage overhead of owners and clients. User revocation is handled securely by the proxy through automatic key regeneration, file re-encryption and targeted key wrapping with a communication overhead of $O(n)$, and the comparison with existing schemes shows lower owner cost, minimal client security risk, denial-proof integrity and seamless dynamic group support.

A key strength of this framework lies in its scalability and adaptability to collaborative environments. Future work will focus on completing the Hyperledger Fabric and IPFS implementation, performing detailed latency benchmarks, and optimizing re-encryption during massive revocations, thereby supporting safer and more scalable decentralized file sharing.

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Appendix

- Reference Paper of the Seminar
  [Title of the base paper on a secure file sharing system based on IPFS and blockchain with decentralized access control and group key management]
- PPT of the Seminar



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